Mandatory deliverable

LAB | Access control in Linux Lab

Skills you will learn

Version Control & Linux: Linux Fundamentals

1. Objective

By the end of this lab, you will be able to:

- Create and manage user groups in Linux.
- Assign and modify file and directory permissions.
- Implement secure access control using Linux permissions.
- Enable group collaboration with shared directories.

2. Tools Required

- **A Linux system** (local or remote VPS).
- **Basic command-line knowledge.**
- **Multiple user accounts for testing access control.**

3. Lab Steps

3.1 Creating a Shared Group

1. **Log in as an Admin User**
   - Ensure you have `sudo` privileges.
2. **Create a new shared group**

```
sudo groupadd sharedgroup
```

Copy Explain

Ask AI Tutor

3. **Add users to the shared group**

```
sudo usermod -aG sharedgroup unix_user
sudo usermod -aG sharedgroup test
```

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4. **Update group ownership for the user's home directory**

```
sudo chgrp sharedgroup /home/unix_user
```

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5. **Ensure changes take effect**

- Log out and log back in for group changes to apply.

3.2 Setting Up the Shared Directory

1. **Modify directory permissions**

```
sudo chmod 770 /home/unix_user
```

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- `770` : Grants full access to the owner and group, denies access to others.

2. **Create the shared file**

```
su - unix_user
touch /home/unix_user/shared_file.txt
```

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3. **Assign group ownership to the shared file**

```
sudo chgrp sharedgroup /home/unix_user/shared_file.txt
```

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4. **Set read and write permissions for the group**

```
sudo chmod 660 /home/unix_user/shared_file.txt
```

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3.3 Verifying Permissions

1. **Check directory permissions**

```
ls -ld /home/unix_user
```

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- Expected output: `drwxrwx---`, confirming correct access settings.

2. **Check file permissions**

```
ls -l /home/unix_user/shared_file.txt
```

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- Expected output: `-rw-rw----`, ensuring group members can edit the file.

3.4 Access and Modify the Shared File

1. **User `unix_user` adds content to the file**

```
echo "Initial content by unix_user." > /home/unix_user/shared_file.txt
```

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2. **User `test` appends content**

```
su - test
echo "Content added by test user." >> /home/unix_user/shared_file.txt
```

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3. **Verify the changes**

```
cat /home/unix_user/shared_file.txt
```

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- Expected output:

```
Initial content by unix_user.
Content added by test user.
```

4. Advanced Access Control (Optional)

4.1 Using `setgid` for Group Collaboration

1. **Ensure new files inherit group ownership**

```
sudo chmod g+s /home/unix_user
```

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2. **Verify the `setgid` bit**

```
ls -ld /home/unix_user
```

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- Expected output: `drwxrws---`, indicating `s` in the group permissions.

4.2 Applying the Sticky Bit for Secure Collaboration

1. **Prevent users from deleting each other's files**

```
sudo chmod +t /home/unix_user
```

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2. **Verify the Sticky Bit**

```
ls -ld /home/unix_user
```

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- Expected output: `drwxrwxrwt`, indicating `t` in the other permissions.

5. Submission Guidelines

1. Take **screenshots** of:
   - Creating and managing user groups.
   - Assigning and modifying file permissions.
   - Verifying access control settings.
   - Testing shared directory and file access.
2. Compile the screenshots into a **Google Doc or PDF**.
3. Upload to **Google Drive** and share the link.
4. Ensure the instructor has **viewing permissions**.

🎉 **Congratulations! You've successfully configured access control in Linux!** 🚀

Lab completed!

Submitted on Sept 18, 2026 a las 15:54

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Edit lab

You have 36 units left to complete

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Skills learned +42% average

Skill: Version Control & Linux +42% 42%

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